Topic: Undetermined coefficients for nonhomogeneous equations

Question: Find the general solution to the differential equation.

$$y'' + 3y' = 2x - 6e^{-3x}$$

Answer choices:

A $y(x) = c_1 + c_2e^{-3x} + \frac{1}{3}x^2 + \frac{2}{9}x + 2xe^{-3x}$

B $y(x) = c_1 + c_2e^{-3x} + \frac{1}{3}x^2 - \frac{2}{9}x + 2xe^{-3x}$

C $y(x) = c_1 + c_2e^{-3x} + \frac{1}{3}x^2 + \frac{2}{9}x + 3xe^{-3x}$

D $y(x) = c_1 + c_2e^{-3x} + \frac{1}{3}x^2 - \frac{2}{9}x + 3xe^{-3x}$

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Solution: B

We'll replace $g(x) = 2x - 6e^{-3x}$ with $g(x) = 0$ to change the nonhomogeneous equation into a homogeneous equation

$$y'' + 3y' = 0$$

then we'll solve the associated characteristic equation.

$$r^2 + 3r = 0$$

$$r(r + 3) = 0$$

$$r = 0, \ -3$$

These are distinct real roots, so the complementary solution will be

$$y_c(x) = c_1 e^{r_1 x} + c_2 e^{r_2 x}$$

$$y_c(x) = c_1 e^{(0)x} + c_2 e^{-3x}$$

$$y_c(x) = c_1(1) + c_2 e^{-3x}$$

$$y_c(x) = c_1 + c_2 e^{-3x}$$

For the particular solution, we'll use $Ax + B$ as our guess for the polynomial function, and we'll use Ce^{-3x} as our guess for the exponential function, so the guess will be

$$y_p(x) = Ax + B + Ce^{-3x}$$

Comparing this to the complementary solution, we can see that $c_2 e^{-3x}$ from the complementary solution and Ce^{-3x} from the particular solution are

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overlapping terms, so we'll multiply Ce^{-3x} from the particular solution by x, such that our guess becomes

$$y_p(x) = Ax + B + Cxe^{-3x}$$

If we go forward with this guess, we'll run into trouble when we take its derivatives. We'll lose the constant, and won't be able to equate coefficients. To fix this, we'll need to multiply $Ax + B$ by x as well.

$$y_p(x) = Ax^2 + Bx + Cxe^{-3x}$$

Taking the first and second derivatives of the guess gives

$$y_p'(x) = 2Ax + B + Ce^{-3x} - 3Cxe^{-3x}$$

$$y_p''(x) = 2A - 6Ce^{-3x} + 9Cxe^{-3x}$$

Plug the derivatives into the original differential equation.

$$2A - 6Ce^{-3x} + 9Cxe^{-3x} + 3(2Ax + B + Ce^{-3x} - 3Cxe^{-3x}) = 2x - 6e^{-3x}$$

$$2A - 6Ce^{-3x} + 9Cxe^{-3x} + 6Ax + 3B + 3Ce^{-3x} - 9Cxe^{-3x} = 2x - 6e^{-3x}$$

$$(2A + 3B) + 6Ax - 3Ce^{-3x} = 2x - 6e^{-3x}$$

$$(2A + 3B) + (6A)x + (-3C)e^{-3x} = (2)x + (-6)e^{-3x}$$

Equate coefficients.

$$2A + 3B = 0 \qquad 6A = 2 \qquad -3C = -6$$

$$2\left(\frac{1}{3}\right) + 3B = 0 \qquad A = \frac{1}{3} \qquad C = 2$$

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$$3B = -\frac{2}{3}$$

$$B = -\frac{2}{9}$$

Then the particular solution is

$$y_p(x) = \frac{1}{3}x^2 - \frac{2}{9}x + 2xe^{-3x}$$

and the general solution is

$$y(x) = y_c(x) + y_p(x)$$

$$y(x) = c_1 + c_2e^{-3x} + \frac{1}{3}x^2 - \frac{2}{9}x + 2xe^{-3x}$$

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Topic: Undetermined coefficients for nonhomogeneous equations

Question: Find the general solution to the differential equation.

$$y'' + 6y' + 10y = 10x^2 - 8x - 10$$

Answer choices:

A $y(x) = e^{x}(c_1 \sin(-3x) + c_2 \cos(-3x)) + x^2 - 2x$

B $y(x) = e^{-3x}(c_1 \sin x + c_2 \cos x) + x^2 - 2x$

C $y(x) = e^{x}(c_1 \sin(-3x) + c_2 \cos(-3x)) + x^2 + 2x$

D $y(x) = e^{-3x}(c_1 \sin x + c_2 \cos x) - x^2 + 2x$

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Solution: B

We'll replace $g(x) = 10x^2 - 8x - 10$ with $g(x) = 0$ to change the nonhomogeneous equation into a homogeneous equation

$$y'' + 6y' + 10y = 0$$

then we'll solve the associated characteristic equation.

$$r^2 + 6r + 10 = 0$$

$$r = \frac{-6 \pm 2i}{2}$$

$$r = -3 \pm i$$

These are complex roots, so the complementary solution will be

$$y_c(x) = e^{-3x}(c_1 \sin x + c_2 \cos x)$$

For the particular solution, we'll use $Ax^2 + Bx + C$ as our guess for the polynomial function.

$$y_p(x) = Ax^2 + Bx + C$$

Taking the first and second derivatives of the guess gives

$$y_p'(x) = 2Ax + B$$

$$y_p''(x) = 2A$$

Plug the derivatives into the original differential equation.

$$2A + 6(2Ax + B) + 10(Ax^2 + Bx + C) = 10x^2 - 8x - 10$$

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$$2A + 12Ax + 6B + 10Ax^2 + 10Bx + 10C = 10x^2 - 8x - 10$$

$$10Ax^2 + x(12A + 10B) + 2A + 6B + +10C = 10x^2 - 8x - 10$$

Equate coefficients.

$$10A = 10 \qquad 12A + 10B = -8 \qquad 2A + 6B + 10C = -10$$

$$A = 1 \qquad B = -2 \qquad C = 0$$

Then the particular solution is

$$y_p(x) = x^2 - 2x$$

and the general solution is

$$y(x) = y_c(x) + y_p(x)$$

$$y(x) = e^{-3x}(c_1 \sin x + c_2 \cos x) + x^2 - 2x$$

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Topic: Undetermined coefficients for nonhomogeneous equations

Question: Find the general solution to the differential equation.

$$y'' - 4y = 13\cos(3x) - 39\sin(3x)$$

Answer choices:

A $y(x) = c_1 \sin(2x) + c_2 \cos(2x) + 3\sin(3x) - \cos(3x)$

B $y(x) = c_1 e^{2x} + c_2 e^{-2x} + 3\cos(3x) - \sin(3x)$

C $y(x) = c_1 \sin(2x) + c_2 \cos(2x) + 3\cos(3x) - \sin(3x)$

D $y(x) = c_1 e^{2x} + c_2 e^{-2x} + 3\sin(3x) - \cos(3x)$

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Solution: D

We'll replace $g(x) = 13\cos(3x) - 39\sin(3x)$ with $g(x) = 0$ to change the nonhomogeneous equation into a homogeneous equation

$$y'' - 4y = 0$$

then we'll solve the associated characteristic equation.

$$r^2 - 4 = 0$$

$$r = -2, 2$$

These are distinct real roots, so the complementary solution will be

$$y_c(x) = c_1 e^{2x} + c_2 e^{-2x}$$

For the particular solution, we'll use $A\cos(3x) + B\sin(3x)$ as our guess for the particular solution.

$$y_p(x) = A\cos(3x) + B\sin(3x)$$

Taking the first and second derivatives of the guess gives

$$y_p'(x) = -3A\sin(3x) + 3B\cos(3x)$$

$$y_p''(x) = -9A\cos(3x) - 9B\sin(3x)$$

Plug the derivatives into the original differential equation.

$$-9A\cos(3x) - 9B\sin(3x) - 4(A\cos(3x) + B\sin(3x)) = 13\cos(3x) - 39\sin(3x)$$

$$-9A\cos(3x) - 9B\sin(3x) - 4A\cos(3x) - 4B\sin(3x) = 13\cos(3x) - 39\sin(3x)$$

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$$-13A\cos(3x) - 13B\sin(3x) = 13\cos(3x) - 39\sin(3x)$$

Equate coefficients.

$$-13A = 13 \qquad -13B = -39$$

$$A = -1 \qquad B = 3$$

Then the particular solution is

$$y_p(x) = -\cos(3x) + 3\sin(3x)$$

and the general solution is

$$y(x) = y_c(x) + y_p(x)$$

$$y(x) = c_1e^{2x} + c_2e^{-2x} + 3\sin(3x) - \cos(3x)$$

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